Project Presentation: *Sarajevo Apartment Price prediction*

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What will be covered?

1. Introduction
   - What is the problem?
   - What is our solution?
2. Dataset
   - How did we get our data?
   - Overview of the gathered data and cleaning
   - Handling missing values
3. Feature Engineering
   - Feature Selection
   - Feature Creation
   - Data cleaning (Again)
4. Results
   - Best Models
   - Evaluation of the Models
   - Picking the best Model
   - Application
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What is the real-world problem?

In the modern housing market, both **apartment sale prices** and **rental prices** can vary drastically depending on factors such as location, size, number of rooms, building condition, and surrounding amenities.

What is the real-world problem?

In the modern housing market, both **apartment sale prices** and **rental prices** can vary drastically depending on factors such as location, size, number of rooms, building condition, and surrounding amenities.

However, in many listings, especially on local platforms, price information is often missing, incomplete, or inconsistent. This makes it difficult for potential buyers, tenants, and even property owners to assess the true value of an apartment or to set a fair and competitive price.

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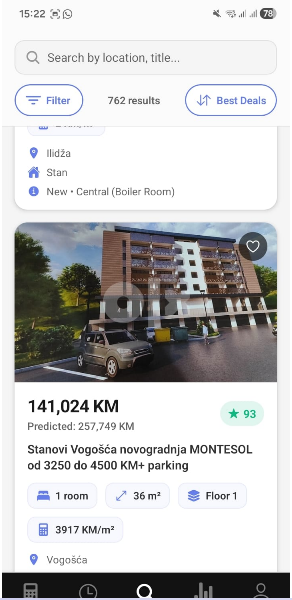
Our Solution

Proposed Approach

- Predict apartment **sale** and **rental** prices
- Use publicly available real-estate data
- Provide transparent price estimates
- Display a “fairness score“ for listings



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How did we get our data?

As of December 30, 2025 there is yet to be any good publicly available data for apartments that were sold or rented out for Bosnia and Herzegovina.

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As of December 30, 2025 there is yet to be any good publicly available data for apartments that were sold or rented out for Bosnia and Herzegovina.

The closest we got to finding public data about apartments in Bosnia and Herzegovina was by using the "Registar cijena nekretnina" that can be found here (*https://www.katastar.ba/rcn/preglednik/*)

The problem with the registry

So there is a public registry of every apartment sales across the federation of Bosnia and Herzegovina.

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The problem with the registry

So there is a public registry of every apartment sales across the federation of Bosnia and Herzegovina.

So can we use it?

Short answer... **no**

Long answer... The data available on the registry, while plentiful, lacks a lot of details needed for us to train a machine learning model

This is why, the only available work done on this topic was just a paper demonstrating the price changes and average prices of real estate sales across the region.

So how did we get our data?

Because of the lack of data available to the public we went with webscraping the data off popular websites for real estate in Bosnia:

- OLX.ba
- Nekretnine.ba

So how did we get our data?

Because of the lack of data available to the public we went with webscraping the data off popular websites for real estate in Bosnia:

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Out of which we were only able to utilize *OLX.ba* as we lacked the time to further pre-process the data from Nekretnine.ba

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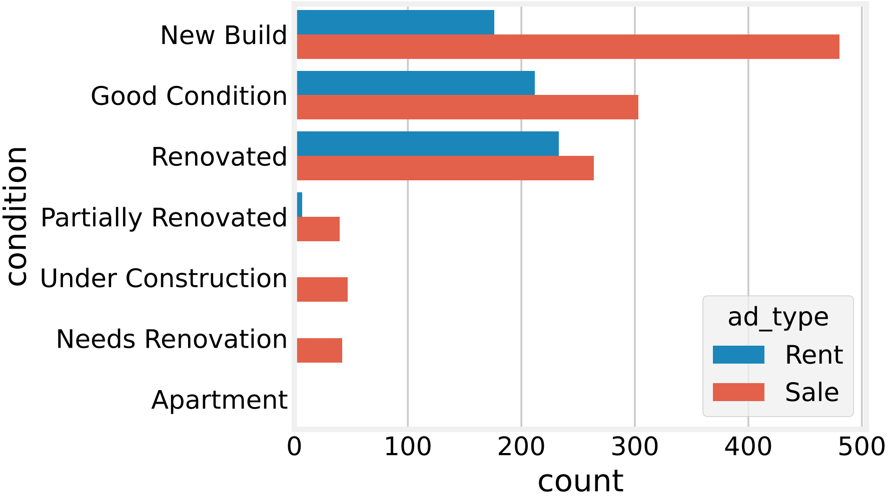
What were we able to gather

The details we were able to collect about every apartment are:

- title
- url
- price
- municipality
- rooms
- m^2
- conditions
- ad type
- property type
- equipment
- level
- heating
- latitude
- longitude

Feature Analysis: Condition



Feature Analysis: Condition

Table: Value counts of condition by ad_type

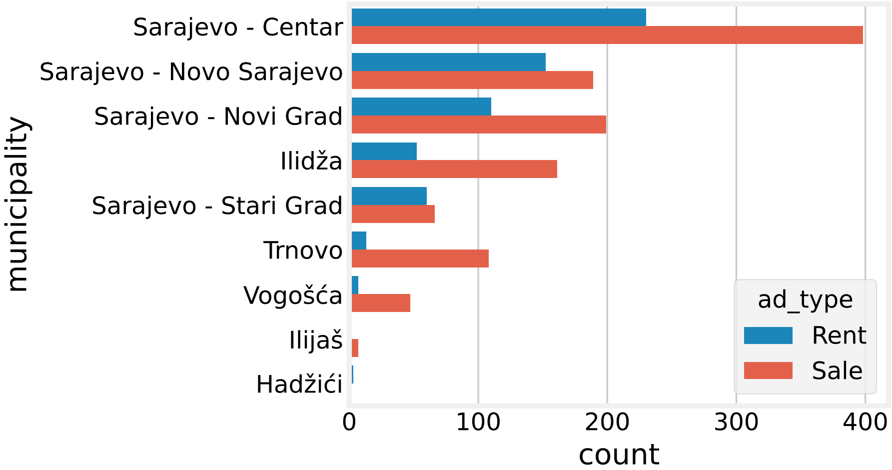
condition	Count
Renovated	233
Good Condition	212
New Build	176
Partially Renovated	7
Needs Renovation	1
Apartment	0
Under Construction	0

(a) Value Counts for Rents

condition	Count
New Build	480
Good Condition	303
Renovated	264
Under Construction	47
Needs Renovation	42
Partially Renovated	40
Apartment	1

(b) Value Counts for Sales

Feature Analysis: Municipality



Feature Analysis: Municipality

Table: Value counts of municipality by ad_type

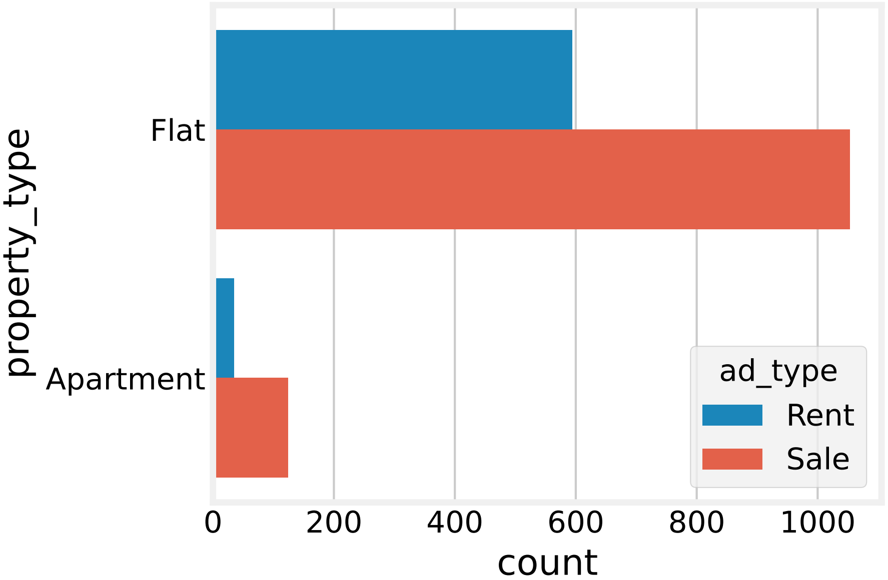
municipality	Count
Sarajevo - Centar	230
Sarajevo - Novo Sarajevo	152
Sarajevo - Novi Grad	110
Sarajevo - Stari Grad	60
Ilidža	52
Trnovo	13
Vogošća	7
Hadžići	3
Ilijaš	2

(a) Value Counts for Rents

municipality	Count
Sarajevo - Centar	398
Sarajevo - Novi Grad	199
Sarajevo - Novo Sarajevo	189
Ilidža	161
Trnovo	108
Sarajevo - Stari Grad	66
Vogošća	47
Ilijaš	7
Hadžići	2

(b) Value Counts for Sales

Feature Analysis: Property Type



Feature Analysis: Property Type

Table: Value counts of property_type by ad_type

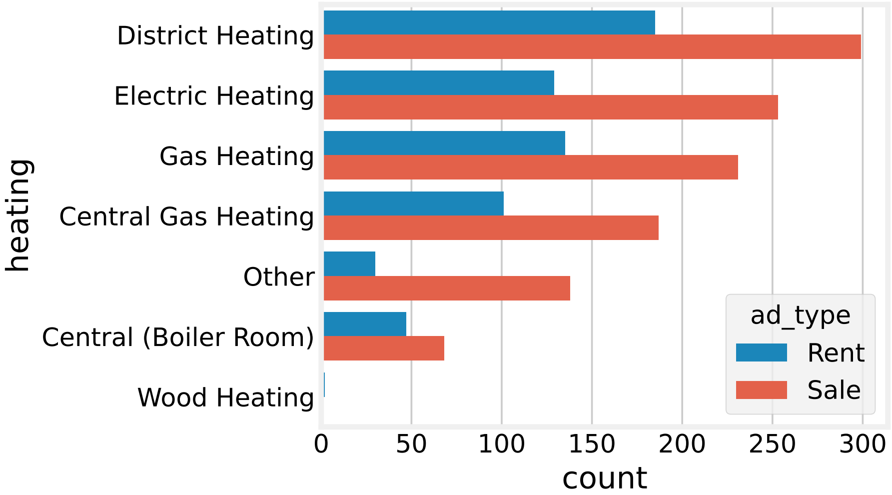
property_type	Count
Flat	594
Apartment	35

(a) Value Counts for Rents

property_type	Count
Flat	1053
Apartment	124

(b) Value Counts for Sales

Feature Analysis: Heating



Feature Analysis: Heating

Table: Value counts of heating by ad_type

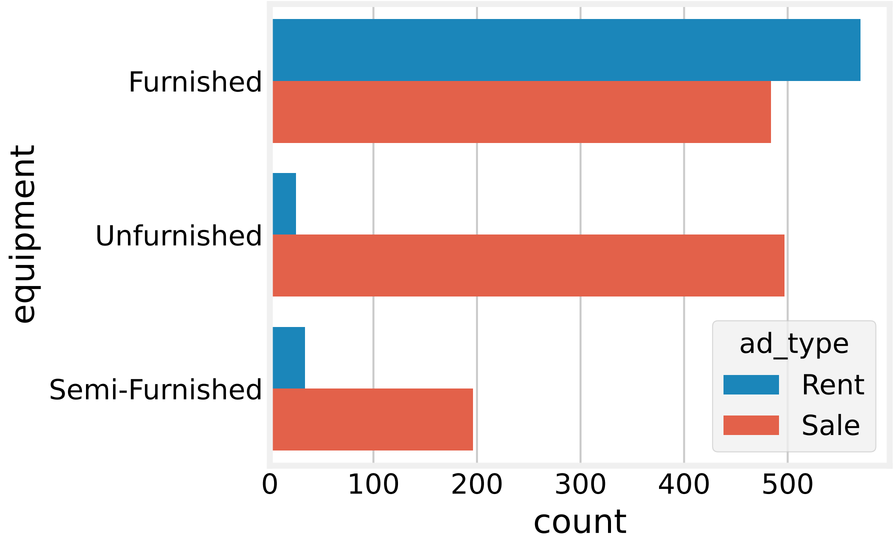
heating	Count
District Heating	185
Gas Heating	135
Electric Heating	129
Central Gas Heating	101
Central (Boiler Room)	47
Other	30
Wood Heating	2

(a) Value Counts for Rents

heating	Count
District Heating	299
Electric Heating	253
Gas Heating	231
Central Gas Heating	187
Other	138
Central (Boiler Room)	68
Wood Heating	1

(b) Value Counts for Sales

Feature Analysis: Equipment



Feature Analysis: Equipment

Table: Value counts of equipment by ad_type

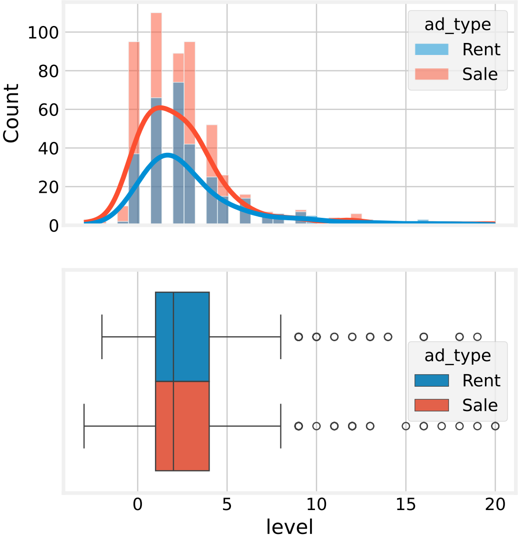
equipment	Count
Furnished	570
Semi-Furnished	34
Unfurnished	25

(a) Value Counts for Rents

equipment	Count
Unfurnished	497
Furnished	484
Semi-Furnished	196

(b) Value Counts for Sales

Feature Analysis: Level



Feature Analysis: Level

Table: Level distribution

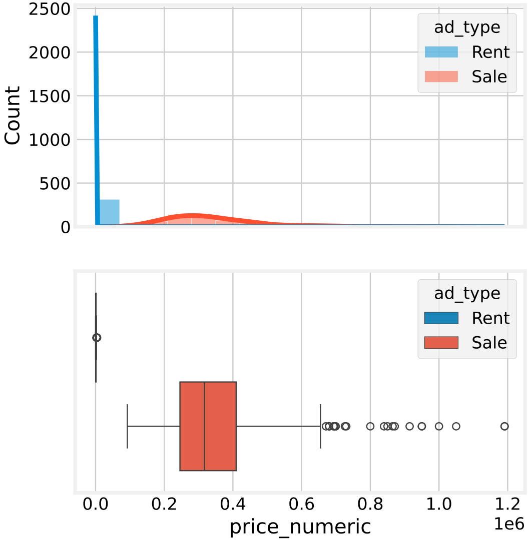
Statistic	Value
count	314.00
mean	3.21
std	3.44
min	-2.00
25%	1.00
50%	2.00
75%	4.00
max	19.00

(a) Level Distribution of Rents

Statistic	Value
count	540.00
mean	2.68
std	3.11
min	-3.00
25%	1.00
50%	2.00
75%	4.00
max	20.00

(b) Level Distribution of Sales

Feature Analysis: Price



Feature Analysis: Price

Table: Price distribution

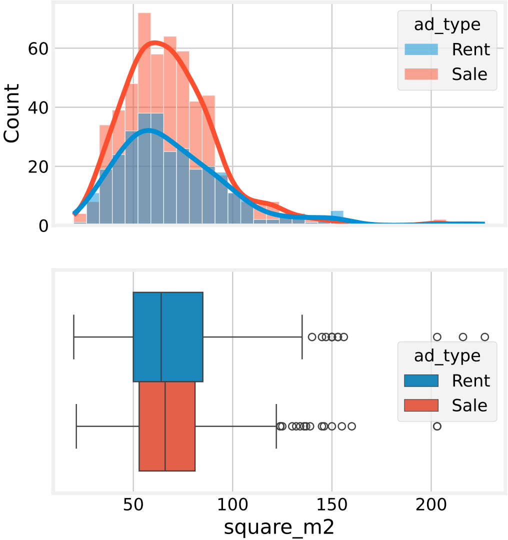
Statistic	Value
count	314.00
mean	1142.35
std	748.54
min	80.00
25%	650.00
50%	900.00
75%	1475.00
max	4500.00

(a) Price Distribution of Rents

Statistic	Value
count	540.00
mean	343392.14
std	152954.51
min	92500.00
25%	245750.00
50%	317000.00
75%	409478.00
max	1191000.00

(b) Price Distribution of Sales

Feature Analysis: Square Meters



Feature Analysis: Square Meters

Table: m^2 distribution

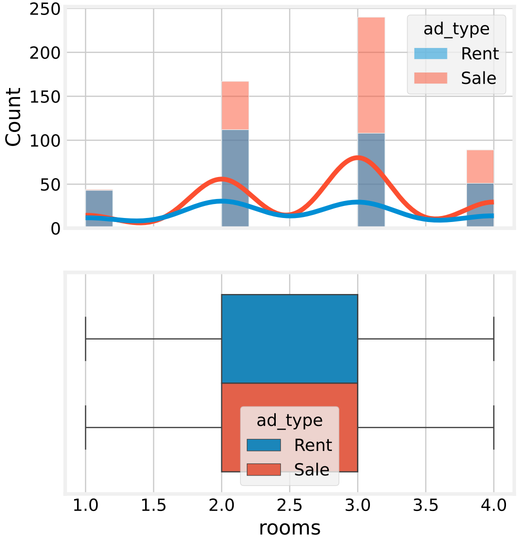
Statistic	Value
count	314.00
mean	70.74
std	30.14
min	20.00
25%	50.00
50%	64.00
75%	85.00
max	227.00

(a) m^2 Distribution of Rents

Statistic	Value
count	540.00
mean	68.63
std	24.38
min	21.30
25%	52.93
50%	66.00
75%	81.08
max	203.00

(b) m^2 Distribution of Sales

Feature Analysis: Rooms



Feature Analysis: Rooms

Table: Rooms distribution

Statistic	Value
count	314.00
mean	2.53
std	0.92
min	1.00
25%	2.00
50%	3.00
75%	3.00
max	4.00

(a) Rooms Distribution of Rents

Statistic	Value
count	540.00
mean	2.69
std	0.84
min	1.00
25%	2.00
50%	3.00
75%	3.00
max	4.00

(b) Rooms Distribution of Sales

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What missing values do we have?

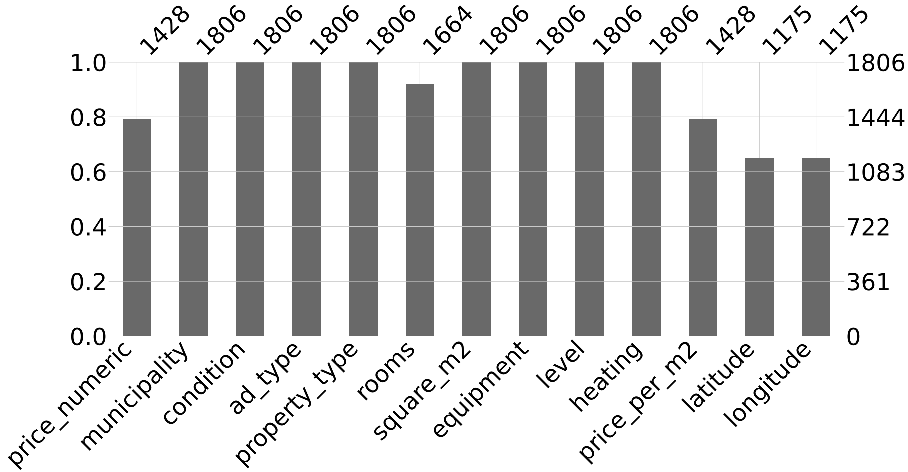

Figure: Bar plot representation of missing features

Any corelation?

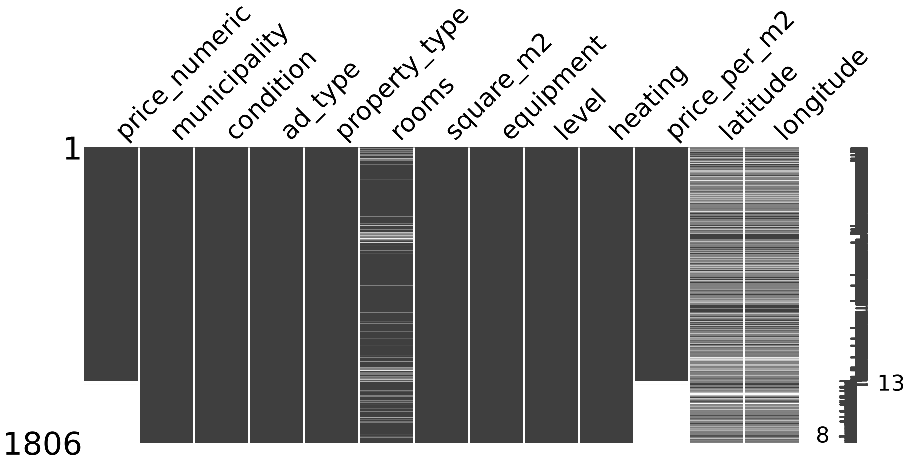

Figure: Matrix plot representation of missing features

Any corelation?

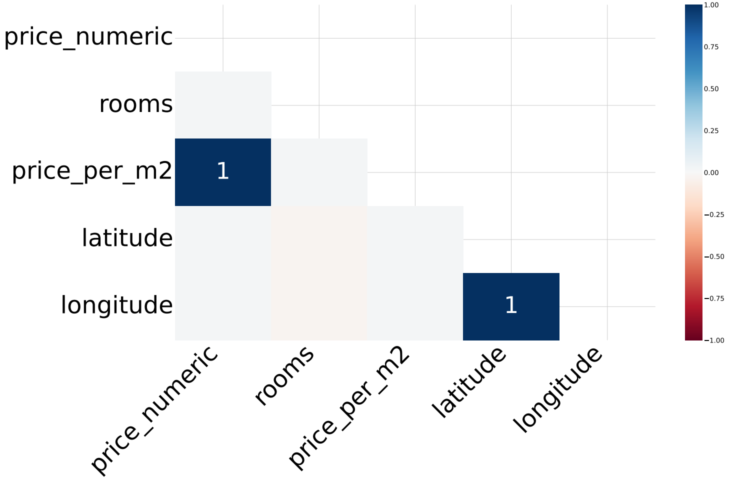

Figure: Heat map of missing feature

Some issues

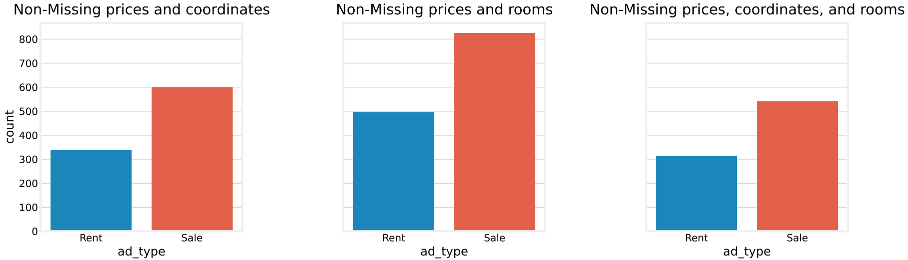

Figure: Combination of missing feature counts

Some issues

If we take a look at all possible combinations of values that we can disregard, we can see that if we get rid of apartments that lack coordinate points, we will lose a lot of data.

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Now the question is, is the loss of these apartments worth it?

In other words, are coordinates of the apartment needed?

Model	CV_R2	Test_R2
GradientBoostingRegressor	0.294	0.314
SVR	0.185	0.270
LinearRegression	0.186	0.266
RandomForest	0.310	0.233
KNeighborsRegressor	0.281	0.197

Table: Score of no coordinates models

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Feature Selection

Removed features

`title`, `url` (non-informative)

`property_type` (high class imbalance, redundant)

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Location-related features

- Municipality alone is insufficient to capture price variation
- Coordinates (latitude, longitude) provide fine-grained location information
- Models trained without coordinates performed poorly ($R^2 \approx 0.27$)

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Final decision

- Keep geographic coordinates
- Exclude municipality from model training
- Use municipality only for stratified train–test splitting

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Additional note

Feature selection also performed automatically during hyperparameter tuning

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Using coordiantes

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One way that we picked to work with coordinates was that first we would load the so-called points of interests of a region, then we would find the closest point to some apartment, calculate the distance from the point to the apartment, and use that as a feature.

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But what is a point of interest?

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One way that we picked to work with coordinates was that first we would load the so-called points of interests of a region, then we would find the closest point to some apartment, calculate the distance from the point to the apartment, and use that as a feature.

But what is a point of interest?

A point of interest can be any amenity that could or would play a role in the pricing of the apartment.

Example of POI

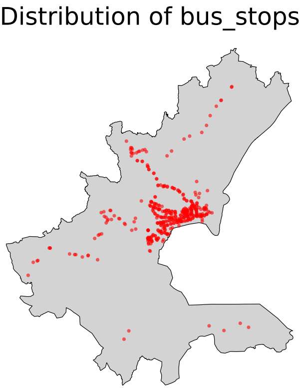

Figure: All bus stops in Canton Sarajevo

Problem

However, using coordinates opens us to another set of problems.

- What if the given apartment was mislabeled?
- What if the apartment was marked as being outside of canton Sarajevo?
- What if the apartment was an agency?

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The idea

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Well there is one simple idea, and that is by checking the location of the apartment and comparing it with its labeled municipality that it claims to be in.

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This is not a perfect approach for removing agencies since agencies can still be in the same municipality that they are selling the apartment. However, the method does work.

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This is not a perfect approach for removing agencies since agencies can still be in the same municipality that they are selling the apartment. However, the method does work.

One nice benefit of this method is that it also lets us remove faulty apartments and in some cases it even helped us remove an apartment that did not exist.

Results

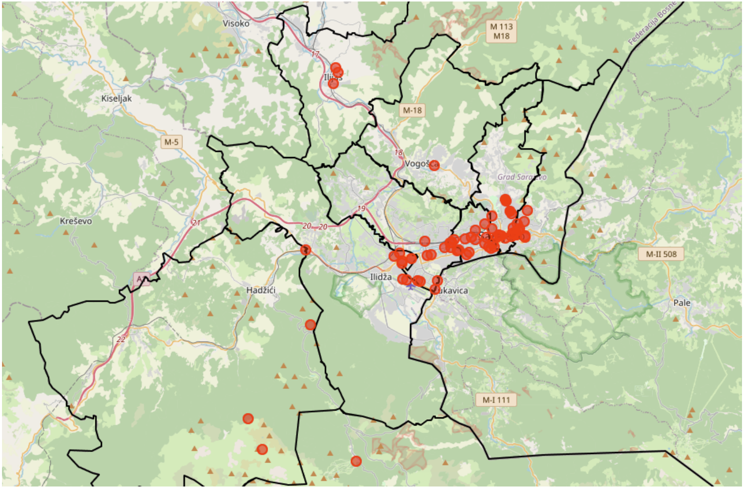

Figure: Removed Apartments

Problem

However, this method does have a problem, and it lies in the fact that human error can happen and usually tend to happen close to the borders.

Problem

Figure: Mislabeled apartment

Problem

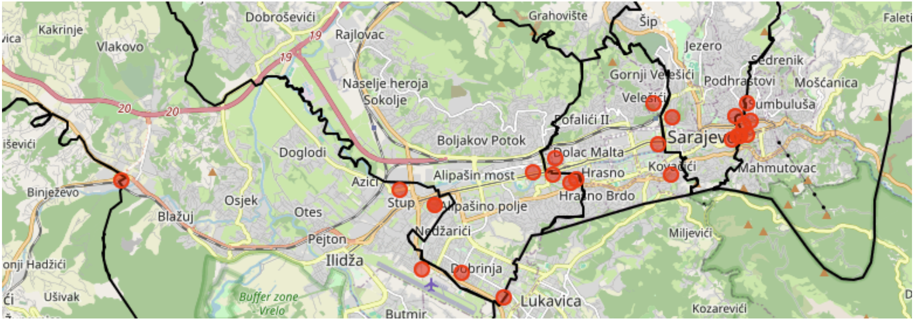

Figure: Corrected Apartments

Final Results

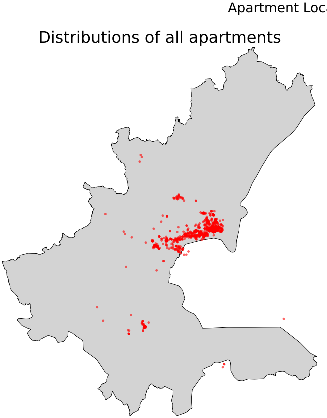

Figure: Comparison of all and corrected apartments

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Linear Regression

Best CV Score	feature_selector	k
0.674	SelectKBest (f-regression)	20.000

Table: Best Linear Regression hyperparameters for apartment sale price prediction

Best CV Score	feature_selector	k
0.421	SelectKBest (f-regression)	30.000

Table: Best Linear Regression hyperparameters for apartment rental price prediction

Random Forest

Best CV Score	max_depth	min_samples_leaf	min_samples_split	n_estimators
0.713	10.000	1.000	2.000	200.000

Table: Best Random Forest hyperparameters for apartment sale price prediction

Best CV Score	max_depth	min_samples_leaf	min_samples_split	n_estimators
0.530	30.000	1.000	2.000	100.000

Table: Best Random Forest hyperparameters for apartment rental price prediction

K-Nearest Neighbors

Best CV Score	algorithm	n_neighbors	p	weights
0.675	ball_tree	5.000	1.000	distance

Table: Best K-Nearest Neighbors hyperparameters for apartment sale price prediction

Best CV Score	algorithm	n_neighbors	p	weights
0.475	ball_tree	5.000	2.000	distance

Table: Best K-Nearest Neighbors hyperparameters for apartment rental price prediction

Support Vector Regression

Best CV Score	C	epsilon	gamma	kernel
0.725	100.000	0.100	scale	rbf

Table: Best Support Vector Regression hyperparameters for apartment sale price prediction

Best CV Score	C	epsilon	gamma	kernel
0.544	100.000	0.100	0.100	rbf

Table: Best Support Vector Regression hyperparameters for apartment rental price prediction

Gradient Boosting

Best CV Score	max_depth	learning_rate	max_iter
0.727	3.000	0.100	400.000

Table: Best Gradient Boosting hyperparameters for apartment sale price prediction

Best CV Score	max_depth	learning_rate	max_iter
0.581	7.000	0.100	400.000

Table: Best Gradient Boosting hyperparameters for apartment rental price prediction

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Evaluation of Sales Models

Model	CV_R2	Test_R2
RandomForest	0.703	0.811
SVR	0.725	0.810
GradientBoostingRegressor	0.727	0.806
LinearRegression	0.674	0.769
KNeighborsRegressor	0.675	0.762

Table: Cross-validation and test R^2 score for sales models

Evaluation of Sales Models

Model	CV_NMAE	Test_NMAE	Baseline_NMAE
RandomForest	-53244.657	-46989.398	-102582.692
SVR	-56652.934	-46456.333	-102582.692
GradientBoostingRegressor	-52631.780	-45336.833	-102582.692
LinearRegression	-61555.003	-55003.266	-102582.692
KNeighborsRegressor	-62187.245	-55960.469	-102582.692

Table: Cross-validation and test Negative Mean Absolute Error score for sales models

Evaluation of Sales Models

Model	CV_NRMSE	Test_NRMSE	Baseline_NRMSE
RandomForest	-83104.333	-63019.613	-145130.067
SVR	-79940.710	-63291.713	-145130.067
GradientBoostingRegressor	-79037.392	-63982.210	-145130.067
LinearRegression	-86950.603	-69762.958	-145130.067
KNeighborsRegressor	-86116.722	-70804.494	-145130.067

Table: Cross-validation and test Negative Root Mean Square Error score for sales models

Evaluation of Rental Models

Model	CV_R2	Test_R2
KNeighborsRegressor	0.475	0.696
RandomForest	0.500	0.672
GradientBoostingRegressor	0.581	0.606
LinearRegression	0.421	0.445
SVR	0.544	0.404

Table: Cross-validation and test R^2 score for rental models

Evaluation of Rental Models

Model	CV_NMAE	Test_NMAE	Baseline_NMAE
KNeighborsRegressor	-362.061	-289.253	-550.323
RandomForest	-327.573	-302.127	-550.323
GradientBoostingRegressor	-320.879	-294.845	-550.323
LinearRegression	-353.536	-331.456	-550.323
SVR	-335.007	-348.715	-550.323

Table: Cross-validation and test Negative Mean Absolute Error score for rental models

Evaluation of Rental Models

Model	CV_NRMSE	Test_NRMSE	Baseline_NRMSE
KNeighborsRegressor	-527.576	-422.298	-766.334
RandomForest	-507.225	-438.776	-766.334
GradientBoostingRegressor	-470.090	-481.294	-766.334
LinearRegression	-550.006	-570.910	-766.334
SVR	-498.346	-591.682	-766.334

Table: Cross-validation and test Negative Root Mean Square Error score for rental models

Visualization of Model: Linear Regression

Sales model

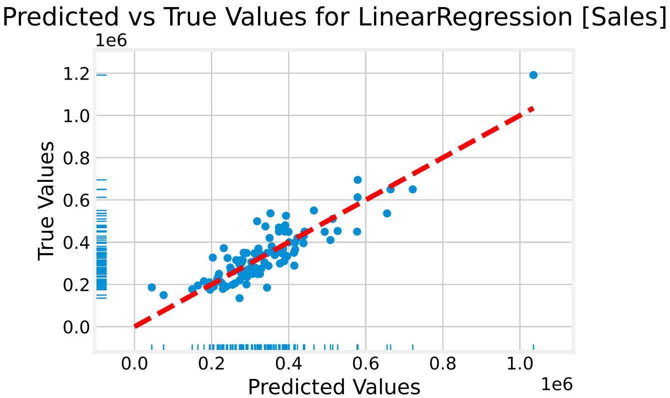

Figure: Predicted versus true apartment sales prices using Linear Regression

Visualization of Model: Linear Regression

Rental model

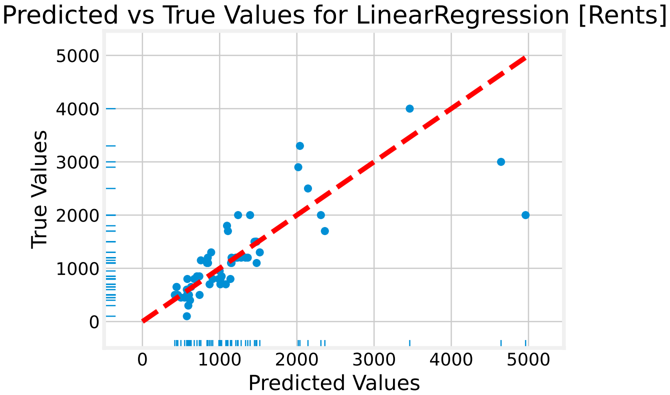

Figure: Predicted versus true apartment rental prices using Linear Regression

Visualization of Model: Support Vector Regression

Sales model

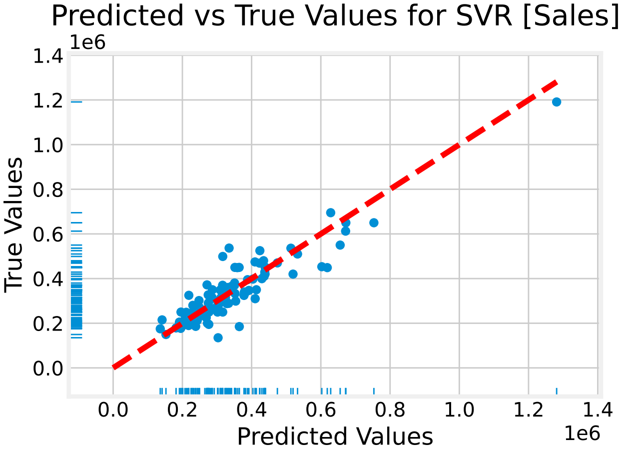

Figure: Predicted versus true apartment sales prices using Support Vector Regression

Visualization of Model: Support Vector Regression

Rental model

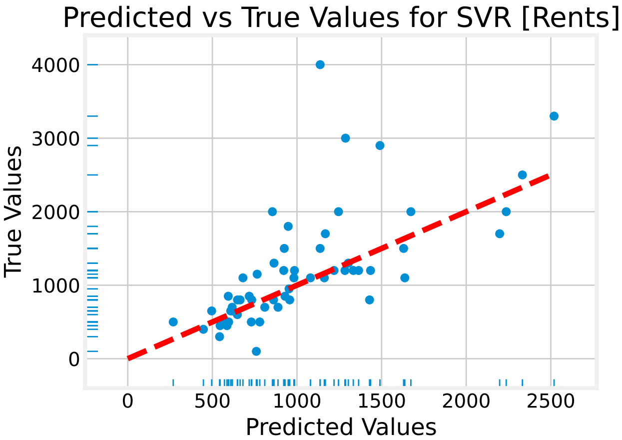

Figure: Predicted versus true apartment rental prices using Support Vector Regression

Sales model

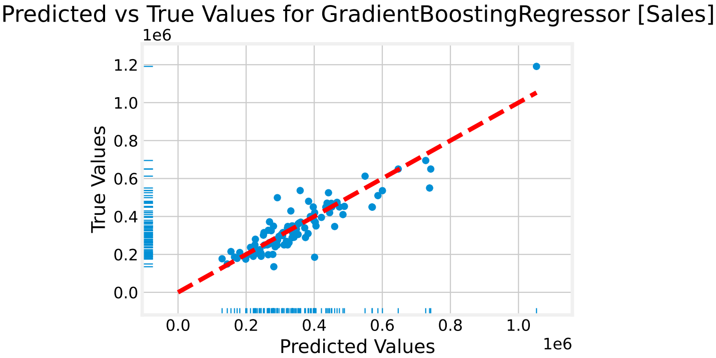

Figure: Predicted versus true apartment sales prices using Gradient Boosting

Visualization of Model: Gradient Boosting

Rental model

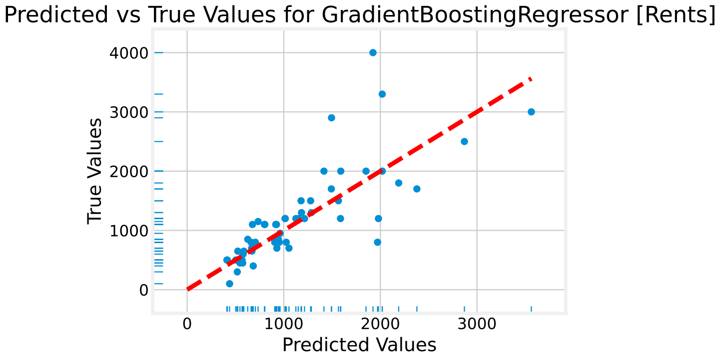

Figure: Predicted versus true apartment rental prices using Gradient Boosting

Visualization of Model: Random Forest

Sales model

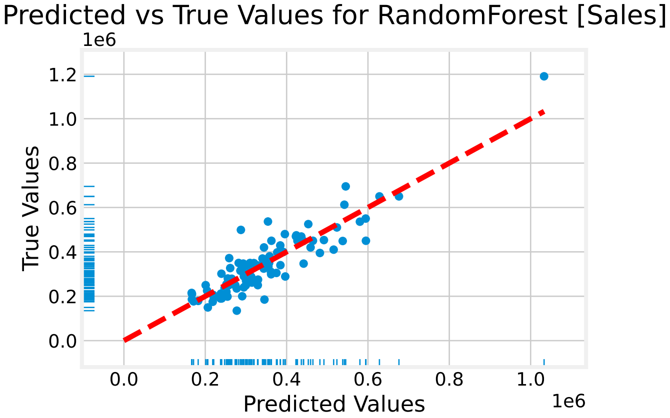

Figure: Predicted versus true apartment sales prices using Random Forest

Visualization of Model: Random Forest

Rental model

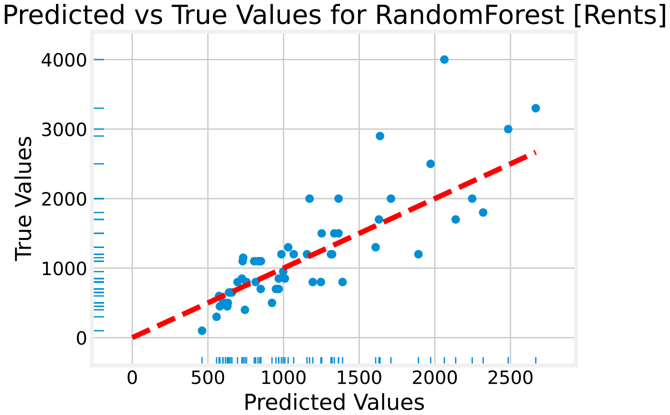

Figure: Predicted versus true apartment rental prices using Random Forest

Sales model

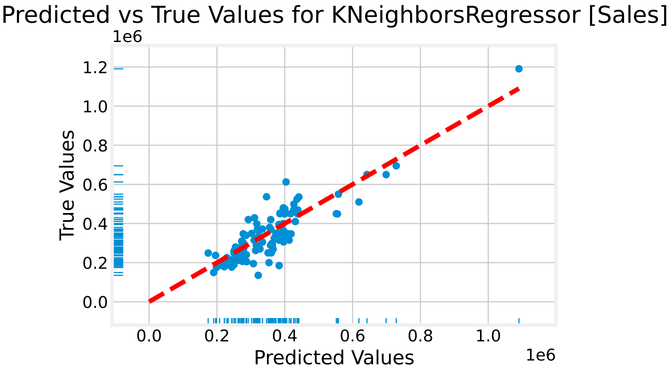

Figure: Predicted versus true apartment sales prices using K-Nearest Neighbors

Visualization of Model: K-Nearest Neighbors

Rental model

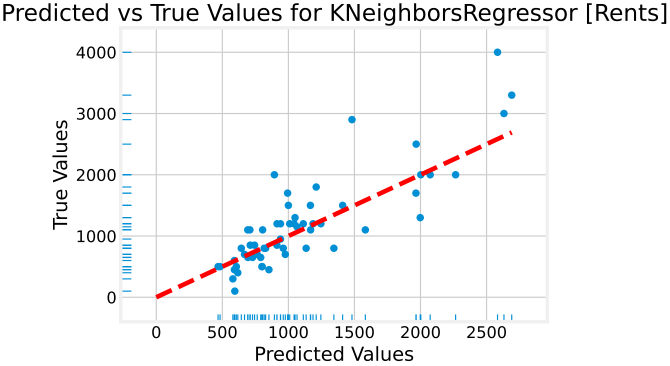

Figure: Predicted versus true apartment rental prices using K-Nearest Neighbors

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Picking the Best Model

Primary criteria

Good generalization on unseen (test) data

Lower test error than the baseline model

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- Lowest test NMAE (average prediction error)
- NRMSE used to resolve close cases

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Final decision

- **Sales model:** Support Vector Regression
- **Rental model:** K-Nearest Neighbors

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Real world example

As mentioned before, we also integrated our machine learning model into an application to give us a nicer feel and to visualize / demonstrate our results

We will now give a quick demonstration of the application

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Contribution Table

Team member name	Specific tasks completed (should not be ambiguous)
Emre Arapčić-Uevak	Exploratory Data Analysis Feature Engineering Machine Learning Performance analysis and result interpretation Preparation of figures, tables, and final report.
Mustafa Sinanović	Web scraping and data collection Assistance with exploratory data analysis Development of a mobile application Integration of the trained machine learning model into the mobile application backend

Table: Work Distribution Table